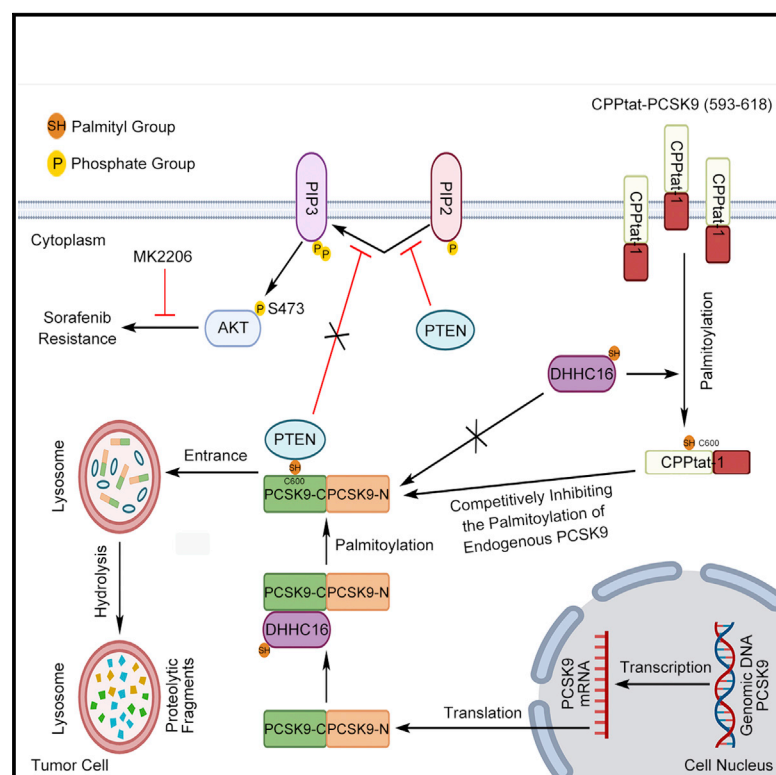


S-palmitoylation of PCSK9 induces sorafenib resistance in liver cancer by activating the PI3K/AKT pathway

Graphical abstract



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In brief

Palmitoylated PCSK9 mediated by ZDHHC16 interacts with PTEN to induce PTEN degradation through lysosome, leading to sorafenib resistance in HCC. However, existing PCSK9 inhibitors fail to improve sorafenib efficacy by preventing PCSK9 palmitoylation. Sun et al. develop a peptide that inhibits PCSK9 palmitoylation to sensitize HCC to sorafenib.

Highlights

- PCSK9 induces sorafenib resistance in HCC by activating PI3K/AKT pathway
- PCSK9 interacts with PTEN, inducing PTEN to enter lysosomes for degradation
- PCSK9 palmitoylation enhances the interaction between PCSK9 and PTEN
- The novel PCSK9 palmitoylation inhibitor improves sorafenib sensitivity in HCC



Article

S-palmitoylation of PCSK9 induces sorafenib resistance in liver cancer by activating the PI3K/AKT pathway

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SUMMARY

Sorafenib is currently the first-line treatment for advanced hepatocellular carcinoma (HCC). However, sorafenib resistance remains a significant challenge. Aberrant AKT signaling activation is a crucial mechanism driving sorafenib resistance in HCC. Proprotein convertase subtilisin/kexin type 9 (PCSK9) plays a vital role in antitumor immune responses. In this study, we demonstrate that aberrant PCSK9 upregulation promotes cell proliferation and sorafenib resistance in HCC by inducing AKT-S473 phosphorylation. After palmitoylation at cysteine 600, the binding affinity between PCSK9 and tensin homolog (PTEN) is dramatically increased, inducing lysosome-mediated PTEN degradation and subsequent AKT activation. We identify zinc finger DHHC-type palmitoyltransferase 16 (ZDHHC16) as a palmitoyltransferase that promotes PCSK9 palmitoylation at cysteine 600. We also develop a biologically active PCSK9-derived peptide that competitively inhibits PCSK9 palmitoylation, suppressing AKT phosphorylation and augmenting the antitumor effects of sorafenib in HCC.

INTRODUCTION

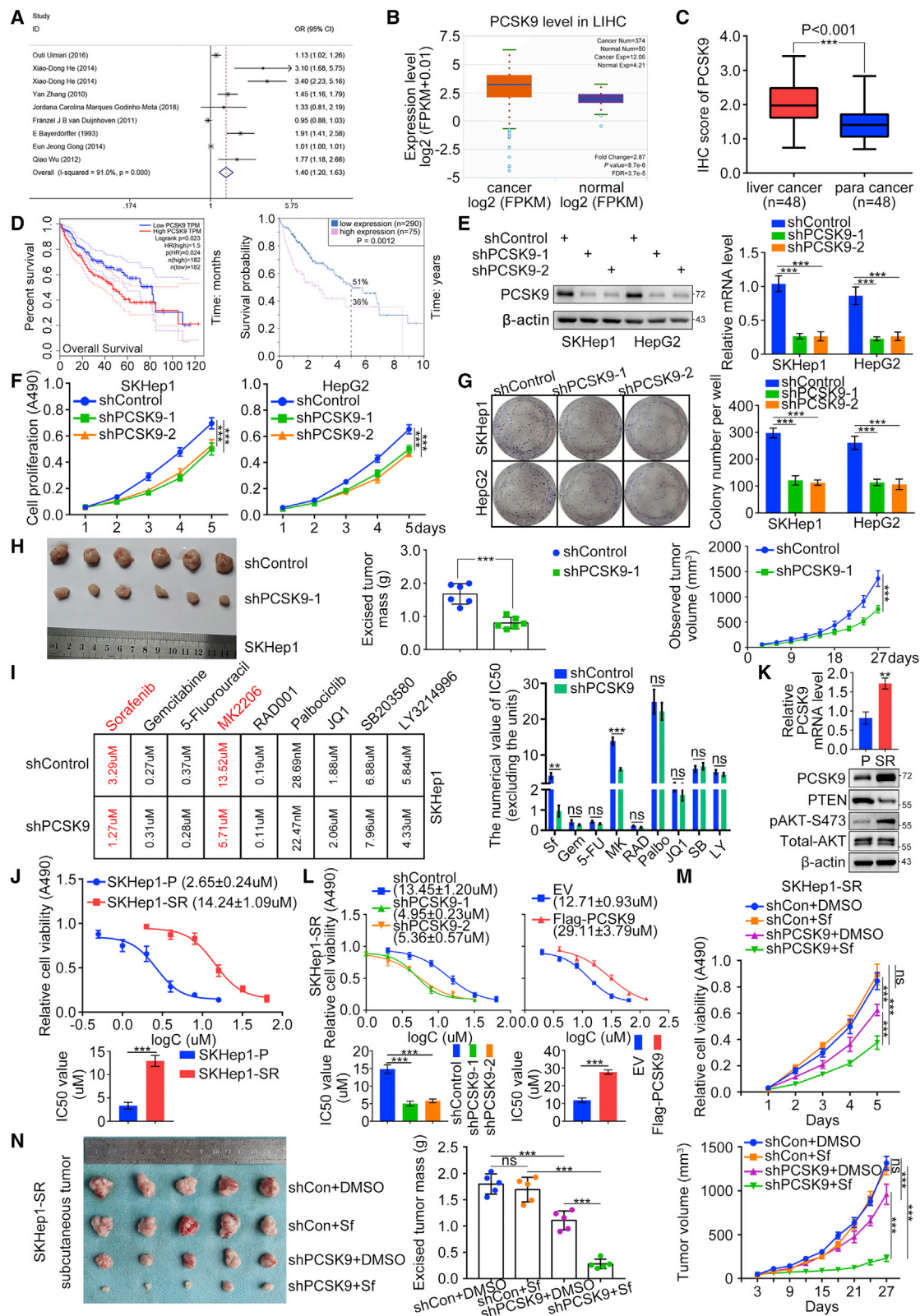
Sorafenib is the first FDA-approved tyrosine kinase inhibitor (TKI) for use in advanced hepatocellular carcinoma (HCC) and is the first-line treatment for patients with HCC (Bruix et al., 2014). Sorafenib can block the activation of mitogen-activated protein kinase (MAPK)/extracellular-signal-regulated kinase (ERK) signaling and suppress cancer cell proliferation by inhibiting the expression of the isoforms of serine/threonine kinases Raf, Raf-1, and B-Raf (Liu et al., 2006). Nonetheless, sorafenib can prolong overall survival (OS) by only 2.3–3 months, failing to achieve the expected clinical effects in patients with HCC (Cheng et al., 2009). Acquired drug resistance is a key cause of the poor efficacy of sorafenib in patients with HCC. Dysregulation of hypoxia-inducible factor-1 (HIF-1) expression (Vaupel et al., 2007), autophagy (Wu et al., 2016), metallothionein (MT) expression (Sun et al., 2016), and phosphatidylinositol 3-kinase (PI3K)/AKT activation (Zhai et al., 2014) has been reported to promote acquired resistance to sorafenib in HCC cells. However, additional mechanisms are thought to promote sorafenib resistance in HCC.

Proprotein convertase subtilisin/kexin type 9 (PCSK9), a key enzyme regulating cholesterol metabolism, binds to low-density

lipoprotein receptor (LDLR) through the PCSK9 catalytic domain to induce LDLR degradation (Zaid et al., 2008). More recently, PCSK9 has also been found to regulate cancer development (Mahboobnia et al., 2021). Specifically, PCSK9 promoted cancer cell proliferation and apoptosis inhibition in neuroglioma (Piao et al., 2015), lung adenocarcinoma (Xu et al., 2017), and melanoma (Sun et al., 2012). Moreover, PCSK9 downregulation increased the levels of major histocompatibility complex (MHC) protein class I molecules on the surface of tumor cells, increasing the infiltration of cytotoxic T cells into tumors and augmenting the efficacy of anti-PD-1 agents (Liu et al., 2020). Although PCSK9 is overexpressed in HCC cells (Zhang et al., 2021), its function in liver cancer remains unclear.

As a lipid phosphatase for phosphatidylinositol 3,4,5-trisphosphate, phosphatase and tensin homolog (PTEN) is recognized as a tumor suppressor by counteracting PI3K/AKT signaling (Cantley, 2002; Song et al., 2012). A PTEN deletion or mutation has been detected in various types of human cancer, including HCC (Chalhoub and Baker, 2009; Luo et al., 2016), and therefore targeted Pten for deletion in hepatocytes has led to the spontaneous development of HCC (specifically, intrahepatic cholangiocarcinoma) in mice (Galicja et al., 2010; Kenerson et al., 2013). In





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addition to PTEN gene mutation, abnormal degradation of the PTEN protein is a cause of liver cancer progression. Multiple studies have shown that PTEN is ubiquitinated by E3 ubiquitin ligase and degraded by the proteasome (Ge et al., 2020; Li et al., 2020; Liu et al., 2018), and the resulting low PTEN levels are insufficient to inhibit HCC progression.

In this study, we found that aberrant overexpression of PCSK9 reduced HCC cell sensitivity to sorafenib, and ZDHHC16 mediated PCSK9 S-palmitoylation. Importantly, PCSK9 interacted with PTEN and induced PTEN degradation through the lysosomal pathway, leading to PI3K/AKT pathway activation in a PCSK9-palmitoylation-dependent manner.

RESULTS

Aberrant PCSK9 expression promotes the proliferation and sorafenib resistance of HCC cells

We analyzed the results of published studies and found that PCSK9 upregulation in certain tumors predicted a poor prognosis (Figure 1A). TIMER dataset revealed elevated expression levels of PCSK9 in liver cancer (Figure 1B), and PCSK9 expression upregulation in both HCC tissues and HCC cell lines (Figures 1C, S1A, and S1B). Aberrant PCSK9 expression was associated with poor OS in patients with HCC (Figure 1D). To investigate the role played by PCSK9 in HCC, we silenced (Figure 1E) or overexpressed (Figure S1C) PCSK9 in SKHep1 and HepG2. MTS and colony-formation assays showed that PCSK9 depletion suppressed HCC cell proliferation *in vitro* (Figures 1F and 1G), and knocking down PCSK9 expression inhibited liver cancer tumor growth *in vivo* (Figure 1H). In contrast, PCSK9 overexpression enhanced HCC progression both *in vivo* and *in vitro* (Figures S1D–S1F). Consistently, immunohistochemical (IHC) staining of mouse tumors revealed that the Ki67 cell proliferation index was decreased after PCSK9 knockdown (KD) (Figure S1G). PCSK9 overexpression led to the opposite effects (Figure S1H).

In addition to studying the role played by PCSK9 in regulating HCC tumor growth, we explored whether PCSK9 affected the drug resistance of liver cancer cells. Therefore, PCSK9 expression was knocked down in SKHep1 cells followed by treatment with an array of small-molecule inhibitors against a number of

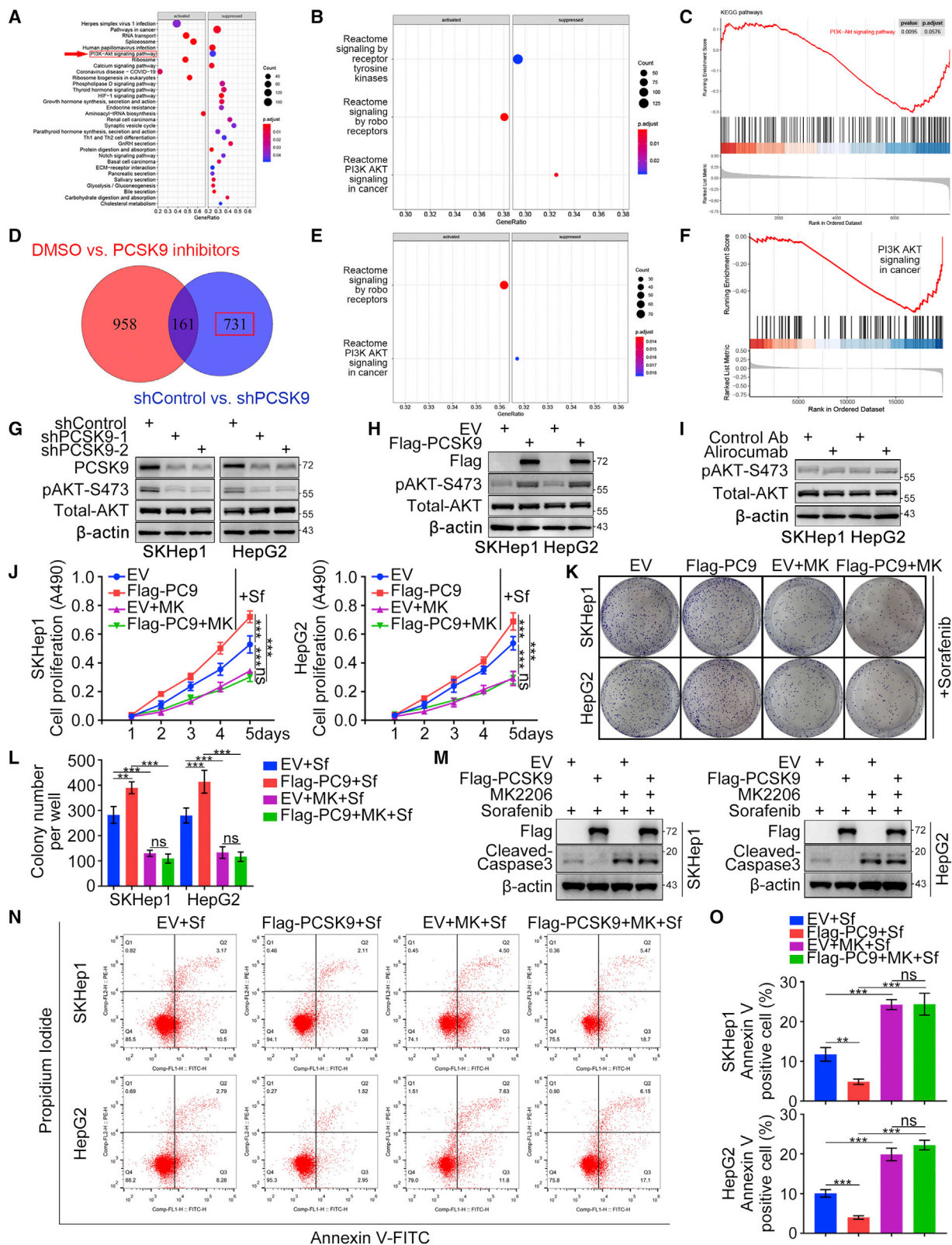
cancer-relevant pathways to determine whether reduced PCSK9 expression affects the sensitivity of liver cancer cells to certain drugs. Interestingly, the drug-sensitivity analyses showed that PCSK9-KD SKHep1 cells were more sensitive to sorafenib and MK2206 after PCSK9 expression was knocked down. In contrast, inhibited PCSK9 expression did not affect the sensitivity of these cells to gemcitabine, 5-fluorouracil, RAD001, palbociclib, JQ1, SB203580, or LY3214996 (Figure 1I). Additionally, sorafenib-resistant SKHep1 (SKHep1-SR) cells were constructed (Figure 1J), and PCSK9 expression was upregulated in SKHep1-SR cells (Figure 1K). Notably, the expression levels of PTEN and pAKT-S473 showed a clear downward trend and upward trend, respectively (Figure 1K). Notably, PCSK9 silencing resensitized HCC cells to sorafenib, while PCSK9 overexpression in HCC cells increased the half maximal inhibitory concentration (IC_{50}) values for sorafenib (Figures 1L and S2A). Moreover, MTS and colony-formation assays showed that PCSK9 silencing enhanced the ability of sorafenib to inhibit HCC cell proliferation (Figures 1M, S2B, and S2C). An Annexin-V/propidium iodide (PI) assay was also performed to provide compelling evidence showing that PCSK9 KD plus sorafenib treatment significantly increased the apoptosis rate (Figure S2D). Additionally, the ability of sorafenib to inhibit HCC tumor growth was significantly increased after PCSK9 expression was knocked down (Figure 1N). In contrast, PCSK9 overexpression decreased the ability of sorafenib to inhibit HCC cell proliferation (Figures 2J–2O, the EV+Sf group versus the FLAG-PCSK9+Sf group). These results suggest that aberrant PCSK9 expression promotes cell proliferation and sorafenib resistance in HCC.

Blocking AKT-S473 phosphorylation improves the PCSK9-mediated sorafenib resistance of HCC cells

We conducted RNA sequencing analysis in SKHep1 cells with PCSK9 knocked down or in cells treated with PCSK9 inhibitors to explore the mechanisms underlying PCSK9 oncogenic functions. A KEGG enrichment analysis showed that the PI3K-AKT pathway was suppressed after PCSK9 KD (Figures 2A–2C). We also identified 161 differentially expressed genes (DEGs) that were common to cells with PCSK9 knocked down and cells treated with alirocumab, and we identified 731 DEGs to cells with

Figure 1. Aberrant PCSK9 expression promotes the proliferation and sorafenib resistance of HCC cells

- (A) A meta-analysis of overall survival in cancers with different PCSK9 levels.
(B) The comparison of PCSK9 levels between HCC and para-carcinoma tissues based on ENCORI Platform.
(C) Boxplot showing the IHC score of PCSK9 proteins in TMA (tissue microarray) ($n = 48$, unpaired t test).
(D) The overall survival of patients with HCC with different PCSK9 levels based on GEPIA and HPA databases.
(E) SKHep1 and HepG2 were transfected with indicated short hairpin RNAs (shRNAs) for 48 and 72 h for qRT-PCR and western blotting, respectively. Two-way ANOVA.
(F and G) Cells were transfected with indicated shRNAs for MTS assay (F) and colony-formation assay (G). $n = 3$, two-way ANOVA.
(H) SKHep1 was infected with lentivirus and injected subcutaneously (s.c.) into the left flank of nude mice. Tumor volumes were measured every 3 days. Tumors were harvested at day 27, photographed, and weighed. $n = 6$, unpaired t test and two-way ANOVA.
(I) SKHep1 was transfected with plasmids for MTS assay to calculate IC_{50} of different drugs. $n = 3$, two-way ANOVA.
(J) IC_{50} value of sorafenib in primary SKHep1 and SKHep1-SR. $n = 3$, unpaired t test.
(K) qRT-PCR and immunoblotting were performed with SKHep1 and SKHep1-SR. $n = 3$, unpaired t test.
(L) SKHep1-SR was transfected with plasmids for MTS assay to measure IC_{50} of sorafenib. $n = 3$, one-way ANOVA and unpaired t test.
(M) SKHep1-SR was transfected with shRNAs for MTS assay after treating with sorafenib ($3 \mu M$). $n = 3$, two-way ANOVA.
(N) SKHep1-SR was infected with lentivirus and injected s.c. into the left flank of nude mice. When tumors reached $50 mm^3$, mice were injected intraperitoneally with sorafenib ($10 mg/kg$) every 3 days. Tumor volumes were measured every 3 days. Tumors were harvested, photographed, and weighed at day 27. $n = 5$, one-way and two-way ANOVA.



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PCSK9 KD (Figure 2D). Interestingly, an analysis of these 731 DEGs confirmed that the PI3K-AKT signaling pathway was suppressed (Figures 2E and 2F), indicating that a PCSK9 inhibitor (alirocumab) might not be necessary to suppress the activation of the PI3K-AKT signaling axis. Consistent with RNA sequencing data, an immunoblot analysis showed that PCSK9 silencing decreased AKT phosphorylation levels in SKHep1 and HepG2 cells (Figure 2G). In contrast, PCSK9 overexpression increased AKT phosphorylation levels in HCC cells (Figure 2H). However, alicumab treatment did not affect AKT phosphorylation levels in HCC cells (Figure 2I).

Moreover, MTS, colony-formation, and Annexin-V/PI assays showed that PCSK9 silencing enhanced the antitumor effect of MK2206 in live cancer cells (Figures S3A–S3C). A xenograft assay and IHC staining of Ki67 *in vivo* showed the same results as those obtained through *in vitro* experiments (Figures S3D and S3E). Considering the key role played by AKT signaling in sorafenib resistance (Zhang et al., 2016), we evaluated sorafenib sensitivity after blocking AKT signaling in HCC cells overexpressing PCSK9. MTS and colony-formation assays showed no difference in the proliferative ability of the HCC cells treated with a combination of MK2206 and sorafenib regardless of whether PCSK9 was overexpressed (Figures 2J–2L and S4A–S4D). Similarly, no difference in HCC cell apoptosis was observed after combination treatment with MK2206 and sorafenib (Figures 2M–2O, S4E, and S4F). Although our results suggested no difference in the proliferative ability of HCC cells after combined treatment with MK2206 and sorafenib regardless of PCSK9 overexpression (Figures 2J–2O and S4A–S4F), PCSK9 silencing enhanced the HCC cells sensitivity to MK2206 treatment (Figures S3A–S3E), indicating contradictory results. We speculated that this discrepancy might have been caused by incomplete blocking of AKT phosphorylation, as shown in Figures S3A–S3E. To test this hypothesis, we examined the inhibitory effect of different MK2206 concentrations on AKT phosphorylation. We found that 10 μ M MK2206 only partially inhibited AKT phosphorylation, while 20 μ M MK2206 almost completely inhibited AKT phosphorylation (Figure S4G). Consistent with previous studies (Ding et al., 2022; Yang et al., 2019), sorafenib treatment with 10 μ M MK2206 almost completely inhibited AKT phosphorylation (Figure S4H). In addition, 10 μ M MK2206 treatment significantly inhibited AKT phosphorylation in cells with PCSK9 expression knocked down (Figure S4I). After obtaining these AKT-phosphorylation-blocking results with MK2206 at 20 μ M, we repeated the MTS (Figure S4J) and colony-formation assays (Figure S4K). The results showed that when AKT phosphorylation was completely suppressed,

PCSK9 failed to regulate the sensitivity of HCC cells to MK2206 (Figures S4J and S4K), which provided a reasonable explanation for the aforementioned contradiction. Generally, these data suggest PCSK9 regulation of sorafenib resistance is mediated through the AKT signaling pathway.

PCSK9 promotes PTEN degradation through the lysosomal pathway to enhance AKT-S473 phosphorylation

The level of endogenous PTEN protein was increased after PCSK9 KD and decreased after PCSK9 overexpressing in SKHep1 and HepG2 (Figures 3A and 3B) but had no overt effect on AKT or PTEN mRNA levels (Figures S5A and S5B). In addition, IHC confirmed that PCSK9 levels were positively correlated with pAKT-S473 levels and negatively associated with PTEN levels in HCC tissues (Figures S5C–S5E). Most importantly, we found that the capacity of PCSK9 to regulate AKT phosphorylation was significantly attenuated after PTEN expression was knocked down regardless of whether PCSK9 was silenced or overexpressed (Figures 3C and 3D). In addition, overexpression of PCSK9 failed to promote AKT phosphorylation when exogenous PTEN proteins were introduced into HCC cells (Figure 3E). These data suggest that PCSK9 promotes AKT phosphorylation by decreasing the level of PTEN protein.

Moreover, a protein mass spectrometry analysis after co-immunoprecipitation (coIP) with an anti-PCSK9 antibody led to the identification of a PTEN peptide, suggesting a possible interaction between PCSK9 and PTEN (Figure 3F). coIP revealed that endogenous PCSK9 and PTEN proteins interacted with each other (Figure 3G). In addition, exogenous FLAG-PCSK9 and hemagglutinin-PTEN were introduced into 293T, SKHep1, and HepG2 for coIP, which showed that PCSK9 and PTEN formed a complex with either exogenous or endogenous proteins (Figures 3H, S5F, and S5G). We also constructed PCSK9 deletion constructs (Figure 3I) and performed a GST pull-down assay. We found that the C-terminal domain, but not the N-terminal domain, in PCSK9 interacted with PTEN (Figure 3J). PCSK9 had been previously reported to promote LDLR and MHC protein class I degradation by directing them to lysosomes (Lagace, 2014; Liu et al., 2020). Therefore, we explored whether PCSK9 also promoted PTEN degradation by importing PTEN into lysosomes. Importantly, the ability of PCSK9 to regulate PTEN protein levels was significantly decreased in SKHep1 and HepG2 cells after lysosome function was suppressed with bafilomycin A1 (Figures 3K and 3L). An immunoblot analysis of fractionated cell lysates confirmed that PCSK9 overexpression increased PTEN levels in lysosomes, whereas PCSK9 silencing decreased

Figure 2. Blocking AKT-S473 phosphorylation improves the PCSK9-mediated sorafenib resistance of HCC cells

(A–C) SKHep1 was transfected with plasmids for RNA sequencing analysis.

(D) The commonly changed genes were summarized into a Venn diagram.

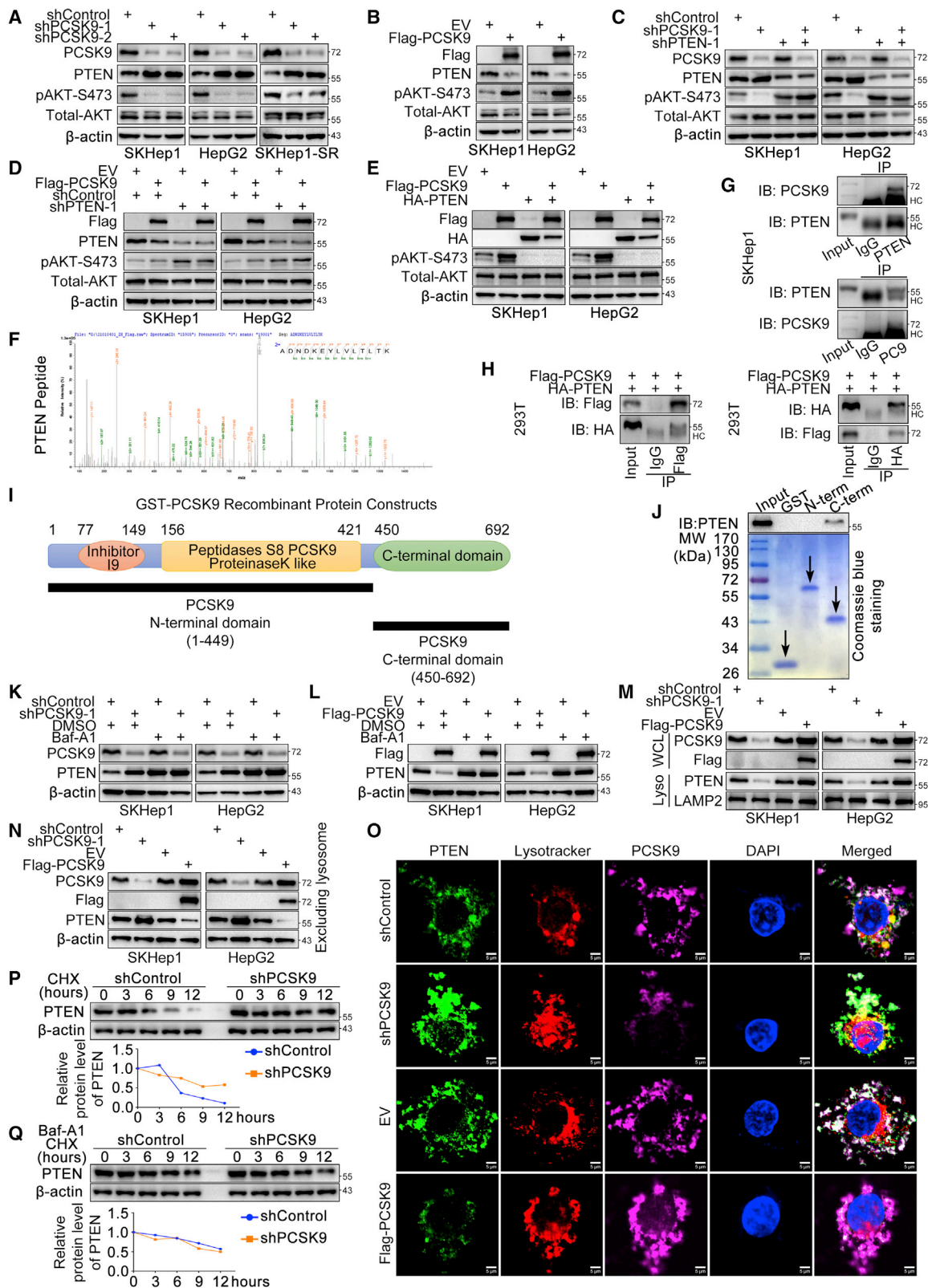
(E and F) KEGG pathway enrichment analysis of shControl versus shPCSK9 after excluding the commonly changed genes.

(G and H) SKHep1 and HepG2 were transfected with shPCSK9 (G) and FLAG-PCSK9 (H) for western blotting.

(I) Cells were treated with control antibody (Ab) or alicumab (10 μ M) for 72 h for western blotting.

(J–L) Cells were transfected with plasmids for MTS (J) and colony-formation assay (K and L) after treating with or without MK2206 (10 μ M) in the presence of sorafenib (3 μ M). n = 3, two-way ANOVA.

(M–O) Cells were transfected with plasmids for 48 h, then cells were treated with or without MK2206 (10 μ M) in the presence of sorafenib (3 μ M) for another 48 h. Cells were harvested for detecting cleaved caspase-3 (M) and fluorescein isothiocyanate (FITC)/PI flow cytometry (N and O). n = 3, one-way ANOVA.



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PTEN levels in the lysosomal fraction (Figure 3M). However, the opposite results were observed with the other cellular fractions (Figure 3N). Additionally, an immunofluorescence assay revealed that the total PTEN level was increased and decreased after PCSK9 KD and overexpression, respectively. This assay also showed lower PTEN levels in the lysosomal fraction after PCSK9 silencing and higher PTEN levels in the lysosomal fraction after PCSK9 was overexpressed (Figure 3O). Bafilomycin-A1 (Baf-A1) and cycloheximide were used to analyze the functional significance of the physical association of PCSK9 and PTEN in lysosomes. In the absence of *de novo* protein synthesis (due to cycloheximide treatment), the PTEN levels in PCSK9-KD cells declined significantly more slowly than those in control cells (Figure 3P). When cells were exposed to Baf-A1 to inhibit lysosome function, the decline in PTEN level was not significant between the control and PCSK9-KD cells in the absence of *de novo* protein synthesis (Figure 3Q). These results suggested that PCSK9 plays a critical role in regulating PTEN levels through lysosome-mediated degradation.

To verify that PCSK9 induces sorafenib resistance in HCC cells by reducing PTEN, we constructed PTEN knockout (PTEN-KO) SKHeP1 and HepG2 cells through the clustered regularly interspaced short palindromic repeat (CRISPR-Cas9) method (Figure S6A). Then, PCSK9 expression was knocked down and overexpressed in PTEN-normal-expressing and PTEN-KO HCC cells, respectively (Figures S6B and S6C). The IC₅₀ valued obtained by MTS assay indicated that sorafenib was no longer affected by a change in PCSK9 expression in SKHeP1 and HepG2 cells with PTEN expression knocked out (Figures S6D and S6E), and MTS and colony-formation assays showed that the cell proliferation rate was not changed in PTEN-KO cells after treatment with sorafenib regardless of whether PCSK9 was overexpressed or knocked out (Figures S6F and S6G). In addition, an Annexin-V/PI assay showed a change in the apoptosis rate of the PTEN-KO cells after treatment with sorafenib (shControl+Sf versus shPCSK9+Sf, EV+Sf versus FLAG-PCSK9+Sf) (Figure S6H). Generally, these results suggest that PCSK9 regulated sorafenib resistance in HCC cells through PTEN.

A previous study has shown that PTEN loss and the subsequent activation of the PI3K/AKT pathway led to cholesterol ester

accumulation in cancer cells (Yue et al., 2014). Therefore, we wondered whether an AKT feedback mechanism underlies PCSK9 regulation of cholesterol metabolism. To test our supposition, different concentrations of an AKT inhibitor (MK2206) and a PI3K inhibitor (GDC-0941) were used to treat SKHeP1 and HepG2 cells. The results showed that the mRNA expression level of PCSK9 was slightly elevated (up to 1.4-fold) with increasing inhibitor concentration, but the change was not statistically significant (Figure S6I). In addition, an immunoblot analysis showed that the expression level of PCSK9 was only slightly elevated (up to 1.4-fold) in SKHeP1 and HepG2 cells after treatment with the AKT and PI3K inhibitors (Figure S6J). In contrast to previous studies (Wu et al., 2021; Xiao et al., 2019), our results did not indicate consistent significant changes in PCSK9 expression.

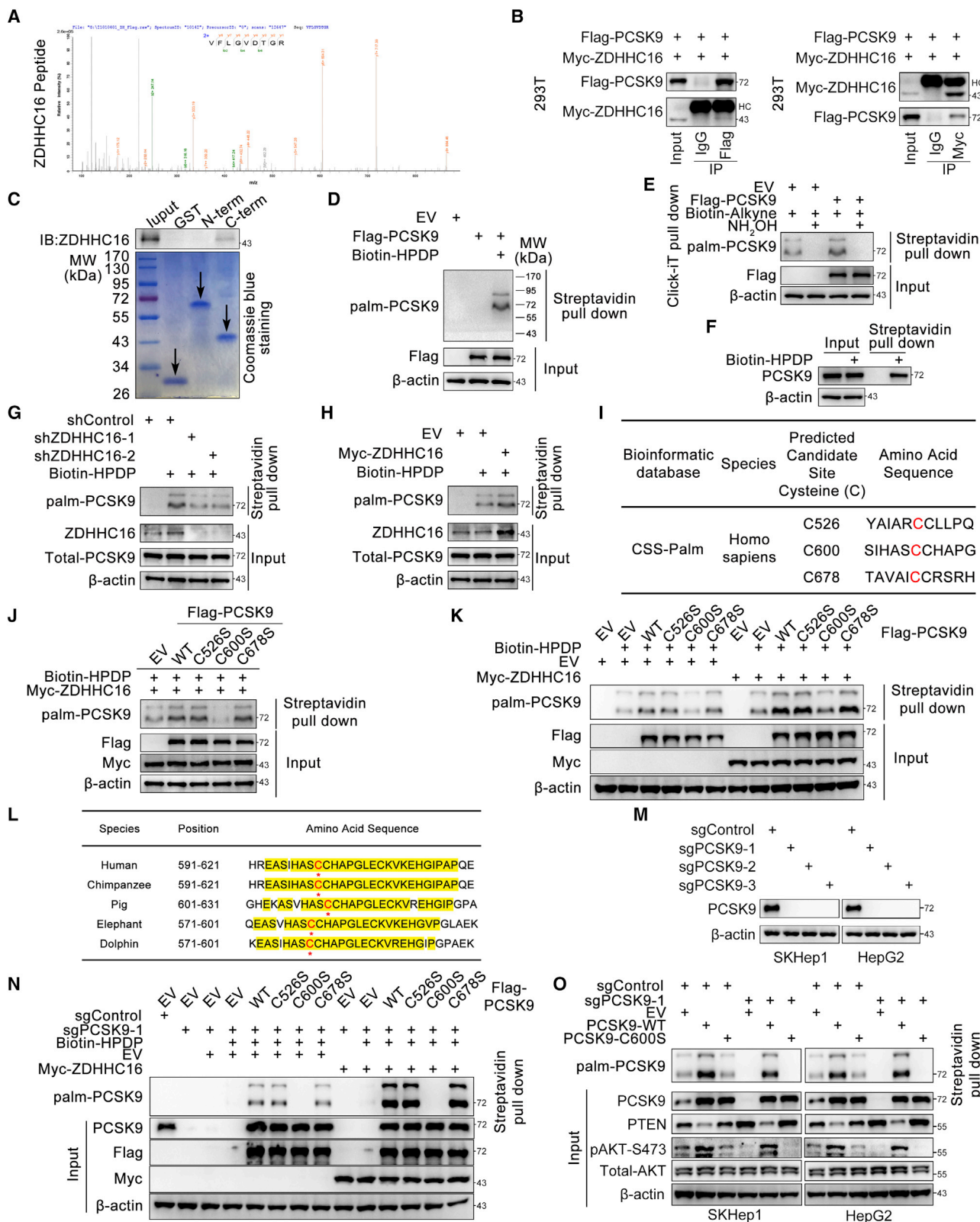
ZDHHC16 is a PCSK9-palmitoylating enzyme that induces PCSK9-C600 S-palmitoylation

Protein mass spectrometry (MS) analysis after CoIP with an anti-PCSK9 antibody detected the ZDHHC16 peptide, which hinted that there was a possible interaction between PCSK9 and ZDHHC16 (Figure 4A). Then, exogenous FLAG-PCSK9 and Myc-ZDHHC16 were introduced into 293T, SKHeP1, and HepG2 cells for coIP, which confirmed that PCSK9 interacted with ZDHHC16 in the presence or absence of exogenous or endogenous proteins (Figures 4B, S7A and S7B). In addition, coIP performed with SKHeP1 demonstrated that endogenous PCSK9 bound ZDHHC16 (Figure S7C). Moreover, PCSK9 deletion constructs were used to explore which PCSK9 domain bound to ZDHHC16. GST pull-down showed that PCSK9 also interacted with ZDHHC16 through its C-terminal domain (Figure 4C).

ZDHHC16 is a palmitoyl acyltransferase regulating the palmitoylation of various proteins (Abrami et al., 2017; Shi et al., 2016). Because PCSK9 interacted with ZDHHC16, we speculated that a PCSK9 cysteine might be palmitoylated by ZDHHC16. Therefore, SKHeP1 expressing FLAG-PCSK9 were used to assess the protein palmitoylation status via the acyl-biotin exchange (ABE) technique, and the analysis revealed that PCSK9 was palmitoylated in HCC cells (Figure 4D). In addition, a Click-iT pull-down revealed that the palmitoylation level of PCSK9 was dramatically

Figure 3. PCSK9 promotes PTEN degradation through the lysosomal pathway to enhance AKT-S473 phosphorylation

- (A and B) SKHeP1 and HepG2 were transfected with plasmids expressing indicated shRNAs (A) and FLAG-PCSK9 (B) for western blotting.
(C–E) Cells were transfected with shRNAs and plasmids for 72 h for western blotting.
(F) PTEN peptide fragment was precipitated with PCSK9 Ab by mass spectrometry (MS).
(G) Western blotting of coIP of endogenous PCSK9 and PTEN in SKHeP1. HC, heavy chain.
(H) 293T was transfected with indicated plasmids. Then, coIP was performed with FLAG or hemagglutinin (HA) antibodies. HC, heavy chain.
(I) The schematic diagram depicting GST-PCSK9 recombinant constructs.
(J) Western blotting of PTEN in SKHeP1 pulled down by GST or GST-PCSK9 (vector: pET-GST). Arrows indicated the protein bands at the expected molecular weight.
(K and L) Cells were transfected with shPCSK9 (K) and FLAG-PCSK9 (L). Then, cells were treated with Baf-A1 (20 nM) for 48 h for western blotting.
(M and N) Cells were transfected with plasmids. Lysosomes were separated from cells. Then, the proteins in and outside lysosomes were extracted for immunoblotting to analyze PTEN levels in the lysosomes (M) and outside the lysosomes (N).
(O) SKHeP1 was cultured in the cell culture slides after transfecting plasmids. Then PCSK9, PTEN, lysosome, and cell nucleus were stained by immunofluorescence technique. Representative images are shown. Scale bar: 5 μ m.
(P) SKHeP1 was transfected with plasmids followed by treatment with CHX (10 μ M). Cells were harvested at different time points to detect PTEN levels.
(Q) SKHeP1 was transfected with plasmids followed by treatment with CHX (10 μ M) and Baf-A1 (20 nM). Cells were harvested at different time points to detect PTEN levels.



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decreased after hydroxylamine treatment (Figure 4E). Immunoblotting for PCSK9 detection also confirmed that endogenous PCSK9 was palmitoylated (Figure 4F). Interestingly, PCSK9 palmitoylation levels were decreased and increased after ZDHHC16 KD and overexpression, respectively, suggesting that ZDHHC16 is a PCSK9-palmitoylating enzyme (Figures 4G and 4H). To identify the cysteine residue(s) in PCSK9 that had been palmitoylated by ZDHHC16, we first utilized CSS-palm software (Ning et al., 2021; Ren et al., 2008) to predict the potential palmitoylation sites, and three possible palmitoylation sites in human PCSK9 were identified (Figure 4I). Hence, we constructed various PCSK9 mutants. However, only the C600S mutation abolished the palmitoylation of PCSK9 (Figure 4J). We also found that the introduction of exogenous Myc-ZDHHC16 promoted the palmitoylation of PCSK9 wild type (PCSK9-WT), PCSK9-C526S, and PCSK9-C678S but not PCSK9-C600S (Figure 4K). Moreover, the C600 palmitoylation site was found to be highly conserved in different animal species (Figure 4L), suggesting that ZDHHC16 is a PCSK9-palmitoylating enzyme that induces S-palmitoylation at PCSK9-C600.

To eliminate the interference of endogenous PCSK9 with exogenous FLAG-PCSK9 protein, SKHeP1 and HepG2 cell lines with PCSK9 expression knocked out (PCSK9-KO) were constructed by the CRISPR-Cas9 method (Figure 4M). Then, exogenous FLAG-PCSK9 plasmids were reintroduced into PCSK9-KO cells to detect the palmitoylation of PCSK9, and the palmitoylation level of the PCSK9-C600S mutant was completely abolished and unaffected by exposure to Myc-ZDHHC16 (Figure 4N). In addition, we found that only PCSK9-WT induced a reduction in the protein expression level of PTEN (Figure 4O). These results suggested that ZDHHC16 promotes the palmitoylation of PCSK9-C600, which is significant for the regulation of PTEN protein levels.

The palmitoylation of PCSK9 dramatically enhances the binding affinity between PCSK9 and PTEN

As PCSK9-WT, but not PCSK9-C600S, was able to obviously reduce the protein level of PTEN (Figure 4O) and ZDHHC16 promoted PCSK9-C600S palmitoylation, we investigated the role of

ZDHHC16 in the ability of PCSK9 to regulate PTEN degradation. Interestingly, PTEN was upregulated upon ZDHHC16 KD in HCC cells. The opposite was observed for pAKT-S473 levels (Figure 5A). However, no changes in PTEN and AKT mRNA levels were observed (Figure S7D). The heatmap of IHC staining revealed that the pAKT-S473 level in HCC tissues with low ZDHHC16 expression was lower than that in HCC tissues with high ZDHHC16 expression, and PTEN showed the opposite trend (Figure S7E), which confirmed that the level of ZDHHC16 was positively correlated with p-AKT and negatively correlated with PTEN (Figure 5B). Representative IHC staining images are shown in Figure S7F.

Notably, ZDHHC16 could no longer regulate PTEN and pAKT-S473 levels in HCC cells with PCSK9 KD (Figure 5C). To this end, we expressed PCSK9-C600S in HCC cells and found that overexpressing PCSK9-C600S failed to increase the palmitoylation level of PCSK9 (Figures 4O and 5D). Furthermore, GST pull-down showed that the interactional intensity was significantly weakened between the GST-PCSK9-C600S mutant and PTEN (Figure 5E), and a coIP assay showed a weaker interaction between PCSK9 and PTEN after suppressing ZDHHC16 expression (Figure 5F). Consistently, immunoblotting also confirmed that PTEN levels remained roughly unaltered in cells expressing PCSK9-C600S compared with the EV (empty vector) group even if Myc-ZDHHC16 was overexpressed (Figure 5G). Immunofluorescence assays also revealed decreased whole PTEN levels and higher PTEN levels in the lysosomal fraction after transfecting PCSK9-WT (Figure 5H). In addition, immunoblotting analysis of fractionated cellular lysates in PCSK9-KO cells also confirmed that PCSK9-WT increased PTEN levels in lysosomes (Figure 5I). These results suggest that the palmitoylation of PCSK9 dramatically enhances the binding affinity between PCSK9 and PTEN, which further promotes PTEN entry into lysosomes for degradation.

Blocking PCSK9-C600 palmitoylation restores sorafenib sensitivity and suppresses HCC progression

Although PCSK9 promoted HCC cell proliferation and sorafenib resistance by activating the AKT pathway, the PCSK9 inhibitor

Figure 4. ZDHHC16 is a PCSK9-palmitoylating enzyme that induces PCSK9-C600 S-palmitoylation

- (A) ZDHHC16 peptide fragment was precipitated with PCSK9 Ab by MS.
- (B) 293T was transfected with indicated plasmids. coIP was performed with FLAG or Myc antibodies. HC, heavy chain.
- (C) Western blotting of ZDHHC16 in SKHeP1 pulled down by GST or GST-PCSK9 (vector: pET-GST). Arrows indicated the protein bands at the expected molecular weight.
- (D) SKHeP1 was transfected with plasmids. Palmitoylated proteins were labeled with biotin-HPDP through acyl-biotinyl exchange (ABE) assay. Then, the palmitoylated proteins were enriched with streptavidin beads for western blotting.
- (E) SKHeP1 was transfected with plasmids. Cells were treated with 100 μ M palmitic acid azide for 6 h. The proteins were reacted with biotin-alkyne and enriched with streptavidin beads. The palmitoylation level of PCSK9 was reduced significantly after treatment with hydroxylamine.
- (F) The endogenous palmitoylated PCSK9 was detected through ABE assay and western blotting in SKHeP1.
- (G and H) SKHeP1 was transfected with shZDHHC16 (G) and Myc-ZDHHC16 (H). Then, PCSK9 was labeled with biotin-HPDP to analyze the palmitoylation levels of PCSK9.
- (I) The palmitoylation sites of PCSK9 were predicted by CSS-Palm software.
- (J and K) SKHeP1 was transfected with plasmids. Then, PCSK9 was labeled with biotin-HPDP for analyzing the palmitoylation levels of PCSK9 by western blotting.
- (L) Alignment of the similarity of PCSK9 sequences in different vertebrate orthologs.
- (M) PCSK9 was knocked out by CRISPR-Cas9 method in SKHeP1 and HepG2.
- (N) SKHeP1 of PCSK9-KO was transfected with plasmids. Then, cells were harvested, and FLAG-PCSK9 was labeled with biotin-HPDP for analyzing the palmitoylation levels of exogenous FLAG-PCSK9.
- (O) SKHeP1 was transfected with plasmids. Then, cells were harvested to analyze the palmitoylation levels of PCSK9 and the levels of PTEN and pAKT-S473.

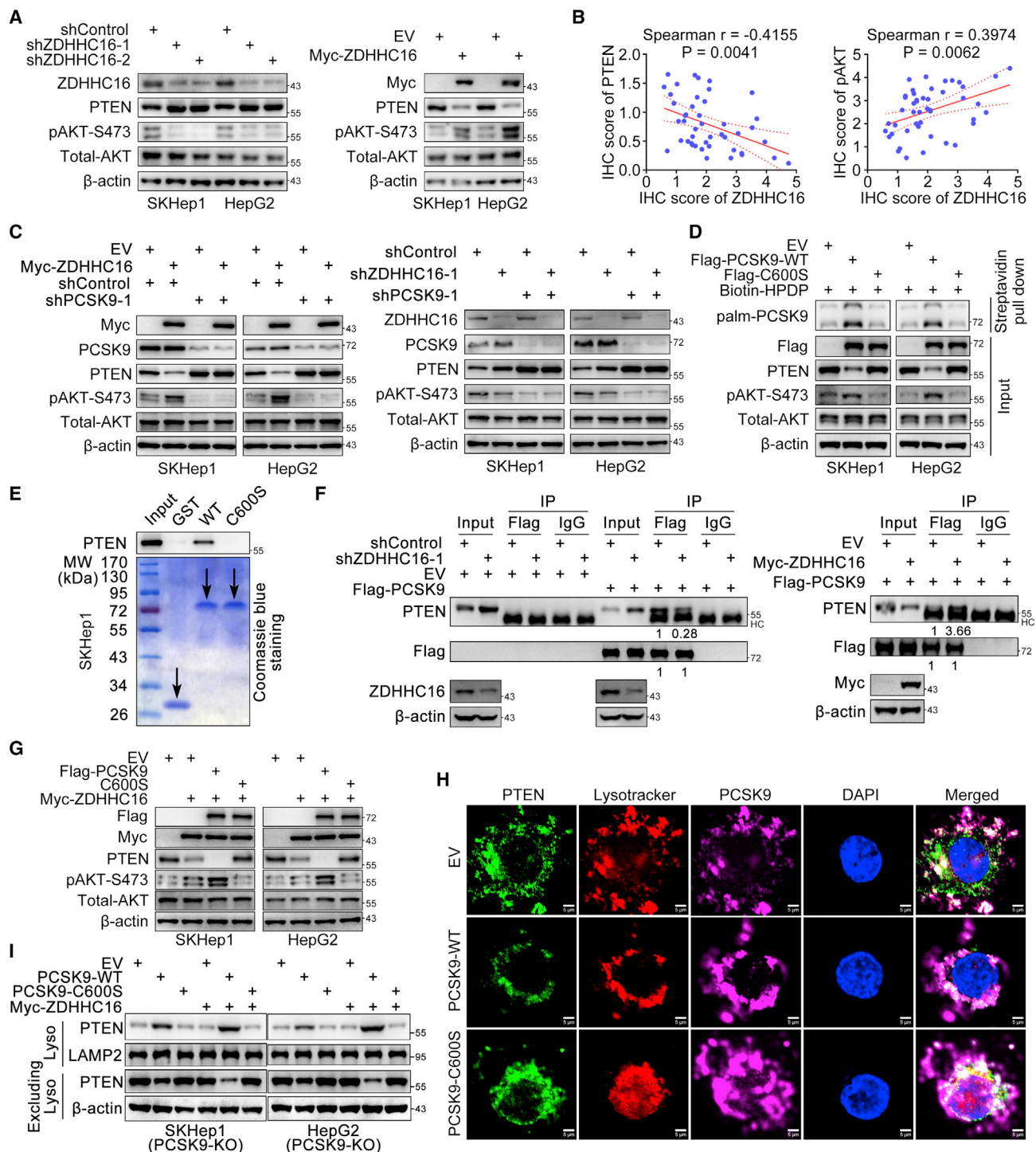


Figure 5. The palmitoylation of PCSK9 dramatically enhances the binding affinity between PCSK9 and PTEN

(A) SKHep1 and HepG2 were transfected with plasmids for 72 h for western blotting.

(B) The correlation analysis of IHC score among ZDHHC16, PTEN, and pAKT-S473.

(C) Cells were transfected with plasmids for 72 h for western blotting.

(D) Cells were transfected with plasmids for 72 h. Then, ABE assays were used to analyze the palmitoylation level of PCSK9.

(E) SKHep1 was transfected with plasmids for 72 h. Then, cells were lysed, and glutathione sepharose beads were used to isolate GST proteins (vector: pcDNA3.1-GST) at 4°C overnight after cells were lysed. Then, GST proteins were eluted with eluting buffer. Western blotting of PTEN proteins pulled down by GST proteins was performed. Arrows indicated the protein bands at the expected molecular weight.

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alirocumab failed to block AKT activation (Figure 2I). This finding may be explained alirocumab binding the catalytic domain and prodomain of PCSK9, not its C terminus (Hess et al., 2018), which led to the inability to prevent PCSK9 palmitoylation. Therefore, an inhibitor to block PCSK9 palmitoylation needed to be developed. On the basis of a peptide designed to target PD-L1 palmitoylation and expression (Yao et al., 2019), green fluorescent protein (GFP) was fused to the PCSK9 sequence (593–618 amino acids) containing the C600 site (GFP-PCSK9-PALM-1) or the mutated C600S site (GFP-PCSK9-PALM-2) to competitively suppress the palmitoylation of endogenous PCSK9 (Figure 6A), and the results of the subsequent assay showed that PALM-1, but not PALM-2, was palmitoylated (Figure 6B). Click-IT pull-down assays showed that PALM-1 palmitoylation not only blocked PCSK9 palmitoylation but also increased the PTEN protein expression level and inhibited AKT phosphorylation, while PALM-2, similar to EV, had no effect (Figure 6C). Moreover, MG132 treatment further increased the PTEN protein expression level in cells transfected with PALM-1. After transfecting PALM-1 and Baf-A1 treatment, the expression of PTEN showed no significant change compared with that after Baf-A1 treatment alone, which indicated that PALM-1 prevented PTEN degradation by lysosomes (Figure 6D). In addition, a coIP assay showed an attenuated interaction between PCSK9 and PTEN due to repressed PCSK9 palmitoylation induced by PALM-1 (Figure 6E). Generally, these results indicated that GFP-PCSK9-PALM-1 suppressed the palmitoylation of endogenous PCSK9, thereby preventing PTEN from being degraded by lysosomes.

A cell-penetrating peptide (CPP) was then fused to PALM-1 (CPPtat-1) and PALM-2 (CPPtat-2) to induce efficient molecular transport (Figure 6F). We found that the degradation rate of PTEN decreased significantly after treatment with CPPtat-1 (Figure 6G), while CPPtat-2 failed to reduce the PTEN-degradation rate (Figure 6H). Moreover, an MTS assay showed that a combination treatment with CPPtat-1 and sorafenib showed a better inhibitory effect on HCC cells than sorafenib treatment alone (Figure 6I). To verify the synergistic effect between CPPtat-1 and sorafenib, sorafenib and CPPtat-1 were simultaneously used to treat SKHep1-SR cells, and the combined application of sorafenib and CPPtat-1 synergistically inhibited AKT phosphorylation (Figure S8A). MTS and colony-formation assays showed that CPPtat-1 increased the sorafenib sensitivity of SKHep1-SR cells (Figures S8B and S8C). The apoptosis rate of the SKHep1-SR cells was also significantly increased after simultaneous treatment with sorafenib and CPPtat-1; specifically, the ability of sorafenib to inhibit tumor cells was enhanced by CPPtat-1 (Figure S8D). Moreover, CPPtat-1 significantly increased the inhibitory effect of sorafenib on HCC cells in mice not only *in vitro* but also *in vivo* (Figure 6J).

We established a mouse model of spontaneously initiated liver cancer through tail intravenous injection of pT3-myr-Akt-HA and pCaggs-NRasV12 plasmids. Representative images of hematoxylin and eosin (H&E)-stained liver tissues are shown in Figure 6K. Compared with normal liver tissues, liver cancer tissues exhibited lower PTEN expression levels, higher p-AKT-S473 levels, and heavier weight (Figures 6K and 6L). Then, CPPtat-1 was injected into the mouse tail vein, and this treatment notably inhibited the progression of liver cancer and reduced the lower pAKT-S473 levels by inhibiting PTEN degradation (Figure 6M). In addition, compared with the effect of the negative control, CPPtat-1 profoundly reduced the mass of the liver cancer tissues (Figure 6N), confirming the antitumor effect of CPPtat-1.

DISCUSSION

PTEN is a tumor suppressor that converts phosphatidylinositol-3,4,5-triphosphate to phosphatidylinositol-4,5-bisphosphate and inhibits the activation of AKT signaling (Myers et al., 1998; Nakakido et al., 2015). PTEN expression and function are altered in multiple tumor types. For instance, the acetylation of lysine 125 and 128 has been shown to inhibit PTEN activity (Okumura et al., 2006). In addition, polyubiquitylation promoted PTEN degradation, whereas monoubiquitylation mediated PTEN nuclear translocation (Trotman et al., 2007; Wang et al., 2007). In addition, many kinases have been reported to phosphorylate PTEN to regulate its stability and function (Al-Khouri et al., 2005). In this study, we found that the levels of PCSK9 and PTEN were negatively correlated and that the interaction between PCSK9 and PTEN led to PTEN degradation and subsequent AKT activation. Moreover, we found that PCSK9 directed PTEN to lysosomes for degradation, similar to the mechanism by which PCSK9 regulates LDLR and MHC I levels via lysosome-mediated degradation. These results suggest a mechanism by which PCSK9 regulates PTEN levels.

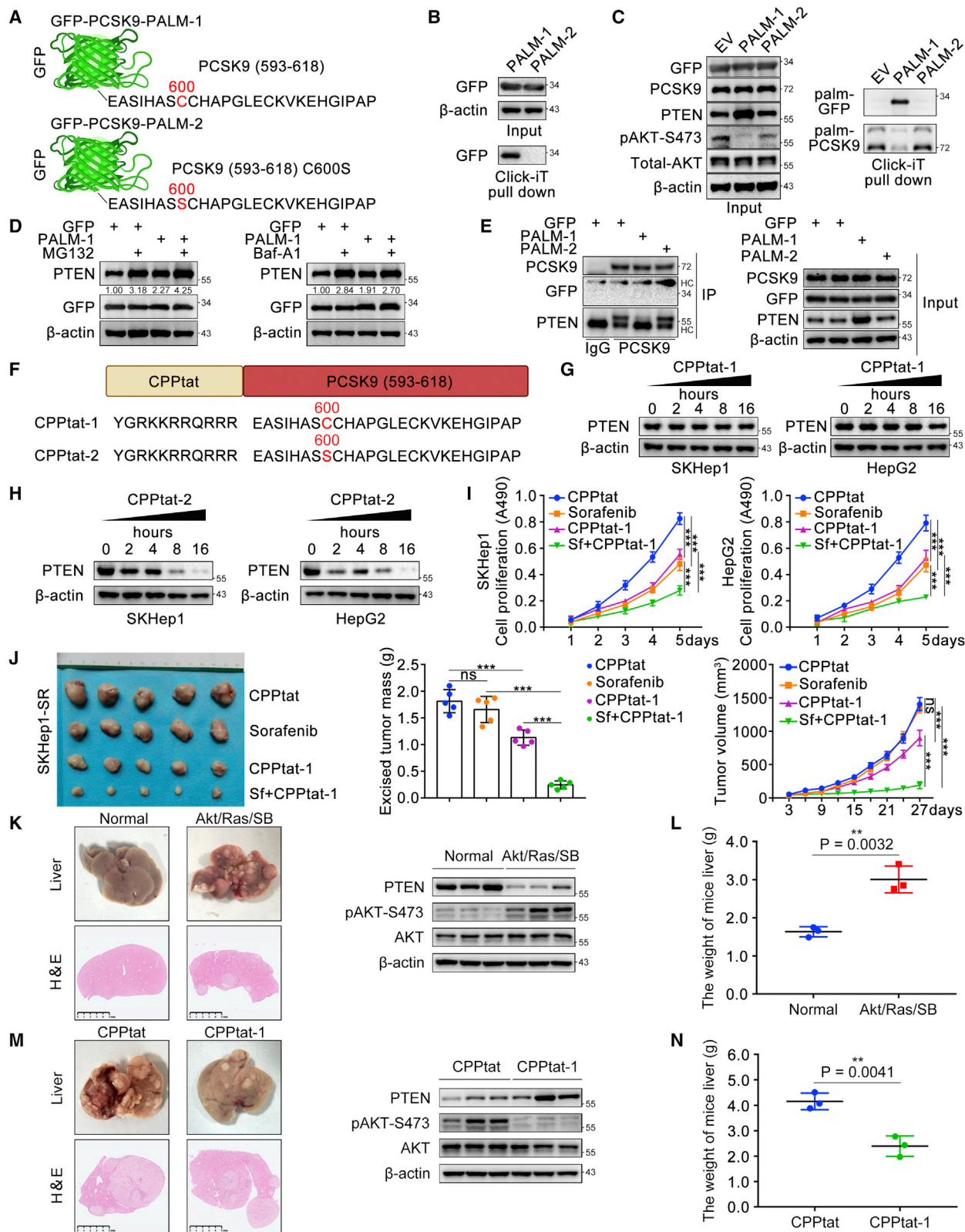
The TKI sorafenib suppresses tumor growth and angiogenesis by targeting multiple kinases (Sun et al., 2016). Although sorafenib has shown potent antitumor effects in patients with advanced HCC, sorafenib resistance is common (Lin et al., 2020). Aberrant activation of PI3K-AKT signaling and PTEN downregulation are important factors driving TKI resistance (Jiang et al., 2018; Morgensztern and McLeod, 2005). Additionally, restoring PTEN expression can significantly increase the sensitivity of tumors to TKIs (Huang et al., 2020; Wang et al., 2018). In this study, we found that PCSK9 silencing in HCC cell lines inhibited AKT phosphorylation and upregulated PTEN expression. Importantly, PCSK9 depletion markedly sensitized HCC cells to sorafenib *in vitro* and *in vivo*. Moreover, the regulatory effect of PCSK9 on sorafenib sensitivity was abolished by treatment with the AKT inhibitor MK2206. These results suggested that aberrant PCSK9 upregulation promotes sorafenib

(F) SKHep1 was transfected with plasmids for 72 h. Then, coIP was performed with FLAG Ab. HC, heavy chain.

(G) SKHep1 and HepG2 were transfected with plasmids for 72 h for western blotting.

(H) SKHep1 was cultured in the cell culture slides after transfecting plasmids. Then, PCSK9, PTEN, lysosome, and cell nucleus were stained by immunofluorescence technique. The representative images are shown. Scale bar: 5 μ m.

(I) SKHep1 and HepG2 of PCSK9-KO were transfected with plasmids. Lysosomes were separated from cells. Then, the proteins in and outside lysosomes were extracted for immunoblotting to analyze PTEN in the lysosomes and outside the lysosomes.



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resistance by promoting AKT phosphorylation in HCC cells. Moreover, we showed that the palmitoylation of PCSK9 in HCC induced PTEN lysosome-mediated degradation, ultimately causing PI3K/AKT signaling activation. These findings are consistent with results from Paul Dent (Hamed et al., 2015) and Ke-Jun Nan (Ruan et al., 2012) et al., emphasizing the critical antitumor roles played by PTEN.

Monoclonal antibodies against PCSK9 have been extensively studied in clinical trials (Hess et al., 2018). However, we found that the anti-PCSK9 antibody alirocumab failed to regulate AKT signaling. Alirocumab has been shown to bind the catalytic domain and prodomain of PCSK9, thereby blocking its interaction with LDLR (Hess et al., 2018; Sarkar et al., 2020), while PCSK9 palmitoylation occurred at the C600 site, not the catalytic domain nor prodomain of PCSK9; therefore, alirocumab failed to inhibit PI3K-AKT signaling pathway activation or attenuate sorafenib resistance. Based on these findings, we developed a biologically active PCSK9-derived peptide that competitively binds endogenous PCSK9 and inhibits its palmitoylation, thereby blocking AKT phosphorylation and enhancing the antitumor effects of sorafenib in HCC.

A previous study showed that PTEN loss and subsequent activation of the PI3K/AKT pathway caused cholesteryl ester accumulation in cancer cells (Yue et al., 2014). PCSK9 is a key enzyme regulating cholesterol metabolism by binding to LDLR (Lagace, 2014). Therefore, we explored whether the AKT signaling pathway regulate the expression of PCSK9 through a feedback mechanism, and we found that the PCSK9 protein expression level was slightly increased (up to 1.4-fold) and that the PCSK9 mRNA expression levels were slightly increased (up to 1.4-fold); neither the protein nor mRNA expression level was significantly elevated (Figures S6I and S6J). Although these results were similar to previously reported data (Wu et al., 2021; Xiao et al., 2019) showing that PI3K/AKT-SREBP2 signaling pathway activation downregulated PCSK9 expression in liver cancer cells, our results indicated that the regulation of PCSK9 expression mediated by the AKT signaling pathway was not obvious. This outcome may be a result of PCSK9 expression directly regulated by SREBP2 because SREBP2 is regulated

by the PI3K/AKT pathway. Hence, PCSK9 was indirectly regulated by the PI3K/AKT pathway in this study.

Limitations of the study

Although we show that palmitoylated PCSK9 mediated by ZDHHC16 interacts with PTEN to promote PTEN degradation through lysosome, leading to sorafenib resistance in HCC, there are still several limitations in this work. First, the reason why PCSK9 was identified as a gene regulating sorafenib sensitivity in HCC was abrupt. SKHep1 cells were measured the IC₅₀ value of several drugs after PCSK9 was knocked down to identify PCSK9 as a gene regulating sorafenib sensitivity, but we did not clarify why the drug sensitivity screening test was performed. In fact, it was more rational and logical to construct a single guide RNA (sgRNA) library to screen PCSK9 as a drug-resistance gene by CRISPR. Second, it was more helpful for our study to initially detect the change of PCSK9 palmitoylation levels after treating with 2-bromopalmitate (2-BP), a pan-inhibitor of PATs. Besides, the current classical protocols for detecting palmitoylation modification are the ABE and Click-iT assays, but there was no way to directly detect palmitoylation modification. Thus, future development of a highly specific palmitoylation-SH antibody could help us further monitor the change of PCSK9 palmitoylation status in HCC. Third, recombinant GST-PCSK9 purified from *E. coli* was able to bind PTEN (Figures 3I and 3J), so could the C600 of PCSK9 also be palmitoylated in *E. coli*? But this issue was not resolved in our study.

Besides, some issues need to be further studied in the future. First, palmitoylation is a reversible modification, and thus palmitoylated PCSK9 has the potential to be dynamic. However, we fail to explore the dynamic balance of PCSK9 palmitoylation and de-palmitoylation levels, which is more important for studying sorafenib resistance in HCC than merely describing PCSK9 palmitoylation. Second, PCSK9 as a secreted protein is matured in the endoplasmic reticulum and then transported to the Golgi, where the protein undergoes posttranslational modifications and is then secreted to an extracellular environment. We fail to explore and explain how mature PCSK9 could escape from the secretory pathway and enter the cytoplasm to interact with

Figure 6. Blocking PCSK9-C600 palmitoylation restores sorafenib sensitivity and suppresses HCC progression

- (A) The green fluorescent protein (GFP) was blended with the PCSK9 sequence (593–618) containing the wild type (GFP-PCSK9-PALM-1) or the C600S mutant (GFP-PCSK9-PALM-2).
- (B) SKHep1 was transfected with plasmids for 72 h. Then, the palmitoylation levels of PALM-1 and PALM-2 were detected by Click-iT chemistry assay.
- (C) SKHep1 was transfected with plasmids for 72 h. Then, the palmitoylation levels of PCSK9, PALM-1, and PALM-2 were detected by Click-iT chemistry assay.
- (D) SKHep1 was transfected with GFP-PCSK9-PALM-1. After 48 h, cells were treating with MG132 (10 μ M) or Baf-A1 (20 nM) for 24 h. Then, cells were harvested for western blotting.
- (E) SKHep1 was transfected with plasmids for 72 h. Then, CoIP was performed with PCSK9 Ab. HC, heavy chain.
- (F) Cell-penetrating peptide (CPP) was blended with the PCSK9 sequence (593–618) containing the wild type (CPPTat-1) or the C600S mutant (CPPTat-2).
- (G and H) SKHep1 and HepG2 were treated with CPPTat-1 (G) or CPPTat-2 (H). Then, the levels of PTEN were detected at different times after treating with CHX (10 μ M).
- (I) SKHep1 and HepG2 were treated with CPPTat-1 (5 μ M) and sorafenib (3 μ M) for MTS assay. $n = 3$, two-way ANOVA.
- (J) SKHep1-SR was injected s.c. into the left flank of nude mice. Mice were treated with or without CPPTat-1 (100 mg/kg) and sorafenib (10 mg/kg) 3 times every 3 days. Tumor volumes were measured every 3 days. Tumors were harvested at day 27, photographed, and weighed. $n = 5$, one-way and two-way ANOVAs.
- (K) The representative images of liver photographs and H&E staining in the spontaneous liver cancer mice. The liver cancer tissues were extracted proteins and subjected to western blotting.
- (L) The comparison of liver quality between normal and spontaneous liver cancer mice. $n = 3$, unpaired t test.
- (M) The representative images of spontaneous liver cancer photographs and H&E staining after treating with CPPTat or CPPTat-1 (100 mg/kg). The liver cancer tissues were extracted proteins and subjected to western blotting.
- (N) The comparison of liver quality between CPPTat and CPPTat-1 treatment groups. $n = 3$, unpaired t test.

and regulate PTEN, which needs to be further studied. Third, although spontaneous liver cancer models of mice provide excellent tools for the study of sorafenib resistance in HCC, the differences between humans and mice hinder the direct clinical translation of novel PCSK9 inhibitor. The clinical studies including organoids and clinical test will reinforce evidence on the impact of this study.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2022.111194>.

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AUTHOR CONTRIBUTIONS

Conception and design, H.W. and X.J.; acquisition of data, Y.S., D.R., and H.Z.; analysis and interpretation of data, X.J., Y.S., and D.R.; writing, review,

and/or revision of the manuscript, X.J. and Y.S.; administrative, technical, or material support, H.Z., F.G., and J.M.; other (technical support), H.W. and X.J.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-PCSK9	Abcam	Cat# ab31762; RRID: AB_777140
Rabbit polyclonal anti-PTEN	Proteintech	Cat# 22034-1-AP; RRID: AB_2878977
Rabbit monoclonal anti-Phospho-AKT (Ser473)	Cell Signaling Technology	Cat# 4060S; RRID: AB_2315049
Rabbit monoclonal anti-AKT	Cell Signaling Technology	Cat# 4685S; RRID: AB_2225340
Rabbit polyclonal anti-Flag	Proteintech	Cat# 20543-1-AP; RRID: AB_11232216
Rabbit polyclonal anti-HA	Proteintech	Cat# 51064-2-AP; RRID: AB_11042321
Rabbit polyclonal anti-LAMP2	Proteintech	Cat# 27823-1-AP; RRID: AB_2880983
Rabbit polyclonal anti-Cleaved Caspase-3	Cell Signaling Technology	Cat# 9661S; RRID: AB_2341188
Rabbit polyclonal anti-ZDHHC16	Sigma-Aldrich	Cat# AV50062; RRID: AB_1858996
Rabbit polyclonal anti-Myc	Proteintech	Cat# 16286-1-AP; RRID: AB_11182162
Rabbit polyclonal anti-GFP	Proteintech	Cat# 50430-2-AP; RRID: AB_11042881
Rabbit polyclonal anti-Beta Actin	Proteintech	Cat# 20536-1-AP; RRID: AB_10700003
Bacterial and virus strains		
E. coli DH5 α	Thermo Fisher	Cat# 18258012
E. coli BL21	Thermo Fisher	Cat# C600003
lentivirus-expressing PCSK9-shRNAs	Sigma-Aldrich	SHCLNG-NM_174936
Chemicals, peptides, and recombinant proteins		
Sorafenib	Selleck	Cat# S7397
Cycloheximide	Selleck	Cat# S7418
Bafilomycin A1	Med Chem Express	Cat# HY-100558
MG132	Med Chem Express	Cat# HY-13259
MK2206	Selleck	Cat# S1078
Gemcitabine	Med Chem Express	Cat# HY-17026
5-Fluorouracil	Med Chem Express	Cat# HY-90006
RAD001	Med Chem Express	Cat# HY-10218
Palbociclib	Med Chem Express	Cat# HY-50767
JQ1	Selleck	Cat# S7110
SB203580	Selleck	Cat# S1076
LY3214996	Selleck	Cat# S8534
GDC-0941	Selleck	Cat# S1065
GFP-PCSK9-PALM-1/2 (recombinant proteins)	Gene chem	N/A
CPPtat-1/2 (recombinant proteins)	Gene chem	N/A
Critical commercial assays		
KOD Plus Mutagenesis Kit	Toyobo	Cat# F0936K
Methyl methanethiosulfonate	Sigma-Aldrich	Cat# 2949-92-0
Hydroxylamine	Sigma-Aldrich	Cat# 7803-49-8
Biotin-HPDP	Top Science	Cat# T19955
Click-iT Protein Reaction Buffer Kit	Thermo Fisher	Cat# C10276
Deposited data		
RNA-Seq data	This paper	GEO: GSE183113
Raw data and images	This paper and Mendeley Data	https://doi.org/10.17632/pcjh4w4jpw.1

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Cell lines		
Human: SKHep1 (a 52-year-old male)	Procell	Cat# CL-0212
Human: HepG2 (a 15-year-old male)	Procell	Cat# CL-0103
Human: HEK293T (embryo)	Procell	Cat# CL-0005
Human: HL7702 (female)	MINGZHOU BIO	Cat# MZ-0625
Human: HuH-7 (male)	Procell	Cat# CL-0120
Human: Bel-7402 (male)	MINGZHOU BIO	Cat# MZ-0633
Human: Hep3B (an 8-year-old male)	Procell	Cat# CL-0102
Experimental models: Organisms/strains		
Mouse: Nude mice (half male and half female, four weeks old)	Vital River	N/A
Mouse: C57BL/6 mice (half male and half female, four weeks old)	Vital River	N/A
Oligonucleotides		
shRNAs, see Table S1	This paper	N/A
Primers for RT-qPCR analysis, see Table S2	This paper	N/A
sgRNAs, see Table S3	This paper	N/A
Recombinant DNA		
Flag-PCSK9	Gene chem	N/A
HA-PTEN	Gene chem	N/A
Myc-ZDHHC16	Gene chem	N/A
Software and algorithms		
ImageJ	NIH	N/A
GraphPad Prism 7.0	GraphPad, Inc	N/A

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Xin Jin (jinxinxy2@csu.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- RNA-seq data have been deposited at GEO and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#). Original western blot images have been deposited at Mendeley and are publicly available as of the date of publication. The DOI is listed in the [key resources table](#). Microscopy data reported in this paper will be shared by the [lead contact](#) upon request.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Cell lines and cell culture

Different human cell lines, HepG2 (a 15-year-old male), SKHep1 (a 52-year-old male), HuH-7 (male), Hep3B (an 8-year-old male) and HEK293T (embryo) were purchased from Procell Life Science&Technology Co., Ltd (Wuhan, China). Human normal hepatocyte HL-7702 (female) and human HCC cell BEL-7402 (male) were purchased from MINGZHOU BIO Co., Ltd (Ningbo, China). All cell lines have been authenticated, and the short tandem repeat (STR) authentication of BEL-7402 cell line and HL-7702 cell line was also provided (Data S1). Cells were cultured in DMEM (Gibco, USA) containing 10% fetal bovine serum (Gibco, USA), 1% Penicillin-Streptomycin (Thermo Fisher Scientific, USA), and 5 μ g/mL plasmocin (InvivoGen, France). Cells were maintained at 37°C in a 5% CO₂ atmosphere.

Establishment of sorafenib-resistant SKHep1 cells

Sorafenib was dissolved in DMSO to a concentration of 1 mM. SKHep1 cells were treated with 3 μ M for two weeks. Subsequently, the medium was replaced with complete DMEM medium, and cells were cultured for another two weeks. This procedure was repeated three times to establish sorafenib-resistant cells (SKHep1-SR).

Mice xenograft model

Nude mice (half male and half female, four weeks old) were purchased from Vital River (China) and were raised on SPF Animal Lab. All mice were divided randomly into groups. Mice were injected subcutaneously with 5×10^6 HCC cells (SKHep1 or SKHep1-SR). When tumors reached 50 mm³, mice were injected intraperitoneally with sorafenib (10 mg/kg), MK2206 (20 mg/kg) or CPPTat-1 (100mg/kg) every three days. The tumor volume was measured every three days. After 27 days, the tumors were harvested, weighed, and photographed. All animal procedures were approved by the Ethics Committee of Tongji Medical College, Huazhong University of Science and Technology.

Spontaneous liver cancer model

C57BL/6 mice (half male and half female, four weeks old) were purchased from Vital River and were raised on SPF Animal Lab. All mice were divided randomly into groups. Plasmids (pT3-myr-Akt-HA:pCaggs-NRasV12:pCMV-SB) dissolved in 2 mL of saline solution were injected into the tail vein of mice (5:25:4 μ g of pT3-myr-Akt-HA:pCaggs-NRasV12:pCMV-SB per 20 g of mouse weight). Tumor formation was assessed by abdominal palpation. Mice with large tumors in the abdomen were injected with CPPTat-1 (100mg/kg) via the tail vein every three days for a total of three injections. Mice were euthanized, and livers were collected after 30 days.

METHOD DETAILS

Lentiviral infections

Lentiviral particles (hU6-MCS-CBh-gcGFP-IRES-puromycin) were used to stably knock down or overexpress PCSK9 in HepG2 and SKHep1 cells. Briefly, 2×10^5 cells were seeded in 6-well plates (day 0). On day 2, the infection medium composed of DMEM and Hitrans G reagent (25:1 ratio) and the appropriate amount of lentiviral particles was added to the cells, and cells were cultivated for another 72 h. Puromycin (10 μ g/mL) was used to select successfully infected cells.

Transient transfection of plasmids

Plasmids were purchased from Shanghai Gene Chem Co., Ltd. Cells were seeded in culture plates or dishes to undergo starvation treatment with 500 μ L Opti-MEM (Gibco, USA) for 12 h. Subsequently, after incubating plasmids (3 μ g) and lipofectamine 2000 (Thermo Fisher Scientific, USA) with 250 μ L Opti-MEM for 5 min, plasmid and lipofectamine mixtures were mixed and incubated for another 20 min. The transfection mixture was added to 500 μ L Opti-MEM, and cells were transfected for 6 h. After transfection, Opti-MEM was replaced with complete DMEM medium. The sequences of all short-hairpin RNAs (shRNAs) are listed in [Table S1](#).

Antibodies and chemicals

The following antibodies were used in this study: anti-PCSK9 (Abcam, ab31762; 1:2000 dilution for WB, 1:200 dilution for IHC), anti-PTEN (ProteinTech, 22034-1-AP; 1:2000 dilution for WB, 1:5000 dilution for IHC), anti-pAKT-S473 (Cell Signaling Technology, 4060S; 1:2000 dilution for WB, 1:2000 dilution for IHC), anti-total-AKT (Cell Signaling Technology, 4685S, 1:2000 dilution for WB), anti-Flag (ProteinTech, 20543-1-AP; 1:2000 dilution for WB), anti-HA (ProteinTech, 51064-2-AP; 1:2000 dilution for WB), anti-LAMP2 (ProteinTech, 27823-1-AP; 1:2000 dilution for WB), anti-cleaved-Caspase 3 (Cell Signaling Technology, 9661S; 1:2000 dilution for WB), anti-ZDHHC16 (Sigma-Aldrich AV50062, 1:1000 dilution for WB; Sigma-Aldrich HPA039893, 1:200 dilution for IHC), anti-Myc (ProteinTech, 16286-1-AP; 1:2000 dilution for WB), anti-GFP (ProteinTech, 50430-2-AP; 1:2000 dilution for WB), and anti- β -actin (ProteinTech, 20536-1-AP; 1:2000 dilution for WB). The following chemicals were used: sorafenib (Multi-kinase inhibitor, Selleck, S7397; 3 μ M *in vitro* and 10 mg/kg *in vivo*), cycloheximide (Protein synthesis inhibitor, Selleck, S7418; 10 μ M), bafilomycin A1 (Lysosomal inhibitor, MedChem Express, HY-100558; 20 nM), MG132 (Proteasome inhibitor, MedChem Express, HY-13259; 10 μ M), MK2206 (AKT1/2/3 inhibitor, Selleck, S1078; 10 μ M *in vitro* and 20 mg/kg *in vivo*), Gemcitabine (DNA synthesis inhibitor, MedChem Express, HY-17026), 5-Fluorouracil (DNA/RNA synthesis inhibitor, MedChem Express, HY-90006), RAD001 (mTOR inhibitor, MedChem Express, HY-10218), Palbociclib (CDK4/6 inhibitor, MedChem Express, HY-50767), JQ1 (BET bromodomain inhibitor, Selleck, S7110), SB203580 (p38 MAPK inhibitor, Selleck, S1076), LY3214996 (ERK1/2 inhibitor, Selleck, S8534) and GDC-0941 (PI3K inhibitor, Selleck, S1065).

Western blot analysis

HCC cells were lysed for 30 min on ice using RIPA lysis buffer (Beyotime, China) with 1% protease inhibitors. The cell lysate was centrifuged at 12000 rpm and 4°C for 10 min, and the supernatant was collected to measure protein concentration using the Micro BCA Protein Assay Kit. The supernatant was boiled at 95°C for 10 min after adding 5 $\times$ loading buffer. Proteins were resolved by Sodium dodecyl-sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and transferred onto PVDF membranes. After blocking with

5% skimmed milk for 1 h at room temperature, the membranes were incubated with the corresponding primary antibodies at 4°C overnight. The next day, the membranes were incubated with the secondary antibodies for 1 h at room temperature, and protein signals were developed using ECL reagent and Bio-Rad Image Lab.

Quantitative real-time PCR (qRT-PCR)

Total RNA was extracted using TRIzol reagent (Thermo Fisher Scientific). RNA concentration and quality were assessed using NanoDrop 2000 (Thermo Fisher Scientific). RNA was reverse-transcribed using PrimeScript RT Kit (Takara, Japan), and TB Green™ Fast qPCR Mix (Takara) was used to amplify cDNA regions of interest. Relative gene expression levels were determined using the 2-ΔCt method after normalizing to β-actin levels. All primer sequences are provided in [Table S2](#).

RNA-seq and data analysis

RNA-seq was performed as previously described ([Jin et al., 2019](#)). Briefly, SKHep1 cells were transfected with shControl or shPCSK9 plasmids for 48 h or treated with DMSO or PCSK9 inhibitor Alirocumab for 48 h. Then total RNA of SKHep1 cells was isolated and extracted from cells using the RNeasy Plus Mini Kit (QIAGEN). High-quality (Agilent Bioanalyzer RIN > 7.0) total RNA was used to prepare the sequencing libraries using the Illumina TruSeq Stranded Total RNA/Ribo-Zero Sample Prep Kit. A total of 500–1000 ng of riboRNA-depleted total RNA was fragmented by RNase III treatment at 37°C for 10–18 min, and RNase III was inactivated at 65°C for 10 min. Size selection (50- to 150-bp fragments) was performed through the FlashPAGE denaturing PAGE-fractionator (Thermo Fisher Scientific) before ethanol precipitation overnight. Then the resulting RNA was directionally ligated, reverse-transcribed and treated with RNase H.

CRISPR/Cas9 technique

Various single guide RNAs (sgRNA) of PCSK9 and PTEN were designed through <https://www.synthego.com>. Then sgRNAs were cloned into lentiCRISPR v2 vector (Addgene, #52961). The sequence of sgRNAs was shown in [Table S3](#).

Co-immunoprecipitation (Co-IP)

HCC cells were lysed in Western/IP lysis buffer (Beyotime, China) on ice for 30 min, and the lysate was centrifuged at 12000 rpm and 4°C for 10 min. The primary antibodies and protein A/G agarose beads (Thermo Fisher Scientific) were added into the supernatant, and samples were incubated overnight at 4°C. The next day, the beads were collected by centrifugation and washed six times with Western/IP lysis buffer. The beads were boiled for 10 min after resuspending in 1× SDS-PAGE loading buffer. BeyoMag™ Anti-Flag Magnetic Beads (Beyotime, P2115), BeyoMag™ Anti-HA Magnetic Beads (Beyotime, P2121) and BeyoMag™ Anti-Myc Magnetic Beads (Beyotime, P2118) were used to precipitate exogenous Flag/HA/Myc-tag proteins, which did not form the heavy chain (HC) to interfere with other bands. The protein identification via mass spectrometry (MS) was performed by SpecAly Life Technology Co., Ltd, Wuhan, China.

Immunohistochemistry

HCC tissue microarray chips (D097LV01) were purchased from Zhongke Guanghai (Xi'an) Intelligent Biological Technology Co., Ltd. The tissue chips were stained with anti-PCSK9, anti-PTEN, anti-pAKT-S473, anti-ZDHHC16 and anti-Ki67 antibodies using an immunohistochemistry (IHC) staining kit (Bios Biological Technology Co., Ltd). IHC scores were independently assessed by two senior pathologists. The staining intensity was defined as (1) weak staining at 100× magnification and little or no staining at 40× magnification; (2) moderate staining at 40× magnification; or (3) strong staining at 40× magnification.

Cell proliferation assay

The MTS assay was used to measure relative cell viability and IC50 values of sorafenib. HCC cells were seeded in 96-well plates (2000 cells in 100 μL DMEM per well), and cells were treated with the appropriate drugs after cell adherence. Subsequently, 20 μL of MTS reagent was added to each well, and cells were incubated for 4 h at 37°C. After adding 200 μL dimethylsulfoxide (DMSO) to each well, the optical absorbance at 490 nm was measured using a microplate reader.

Colony formation assay

HCC cells were seeded in 6-well plates (500 cells in 2 mL DMEM per well). The next day, cells were treated with the appropriate drugs, and the medium was replaced every three days. After two weeks, cells were fixed with 4% paraformaldehyde for 30 min and stained with crystal violet for 20 min.

Glutathione S-transferase (GST) pull-down

Escherichia coli BL21 (expressed strain) was lysed using muramidase and sonication to extract GST-fusion proteins (Vector: pET-GST). Glutathione-Sepharose beads (GE Healthcare Life Sciences, USA) were used to isolate GST-fusion proteins at 4°C overnight. The beads were collected and washed six times. Cells were lysed in Western/IP lysis buffer for 30 min on ice, and the cell lysate was incubated with the beads at 4°C overnight. The beads were boiled for 10 min after washing six times with binding buffer.

Escherichia coli DH5 α (clonal strain) and GST-tag plasmids (Vector: pcDNA3.1-GST) were mixed in ice-bath for 30min, then the mixture was incubated in ice-bath for 2min again after heat shock in 42°C for 90 seconds. The mixture was incubated for 60min on an incubator shaker at 37°C after adding 500 μ L of Luria-Bertani (LB) medium. Then amplified *Escherichia coli* DH5 α was added into 50mL of LB medium containing ampicillin to incubate for 12h on the incubator shaker at 37°C. The next day, the bacteria were harvested and extracted GST-tag plasmids through QIAGEN Endo-Free Plasmid Maxi Kit (QIAGEN, 12362). Then GST-tag plasmids were transfected into HCC cell lines for 72h to perform GST pull-down. In brief, cells were harvested and lysed to extract proteins. Glutathione-Sepharose beads were used to isolate and purify GST-tag proteins at 4°C overnight. The beads were collected and washed for six times with binding buffer. Then the Glutathione-Sepharose beads were boiled for 10min after adding 1xSDS-PAGE loading buffer. Finally, immunoblotting analysis was subjected to analyze the protein-protein interactions.

Flow cytometry

HCC cells were harvested, washed with phosphate-buffered saline (PBS), and resuspended in binding buffer. Cells were incubated with Annexin V-FITC (5 μ L) for 15 min in the dark. After adding 5 μ L of propidium iodide (PI), cells were incubated for another 5 min. Samples were analyzed on BD FACSCelesta (BD Biosciences, USA), and flow cytometry data were analyzed using FlowJo.

Acyl-biotinyl exchange (ABE) assay

HCC cell lysates were incubated with 20 mM of methyl methanethiosulfonate (Sigma) at 50°C for 20 min to block free thiols. Proteins were precipitated with acetone and resuspended in 1 M hydroxylamine pH 7.4 (Sigma) to promote de-palmitoylation. Then, proteins were incubated with 0.2 mM biotin-HPDP (Top Science) for 1 h at room temperature. Streptavidin was used to purify biotinylated proteins, and biotinylated proteins were analyzed by immunoblotting.

Click-iT identification of PCSK9 palmitoylation

HCC cells were transfected with PCSK9 plasmids for 48 h. Subsequently, cells were incubated with 100 μ M of Click-iT palmitic acid-azide for 6 h. After incubation, cells were lysed to extract proteins. Click-iT Protein Reaction Buffer Kit (catalog number C10276; Thermo Fisher Scientific) was used to catalyze the reaction of protein samples with biotin-alkyne. Biotin alkyne-azide-palmitic-protein complex was precipitated with streptavidin, and samples were immunoblotted for PCSK9.

Immunofluorescence assay

HCC cells were incubated with Lyso-Tracker (Thermo Fisher Scientific) for 20 min, fixed with paraformaldehyde for 15 min, and permeabilized with 0.2% Triton X-100 for 10 min. Cells were incubated with anti-PTEN and anti-PCSK9 antibody at 4°C overnight, followed by incubation with fluorescent secondary antibodies for 1 h. Cell nuclei were stained with DAPI for 10 min, and samples were analyzed by confocal microscopy (Andor, Dragonfly, 63 $\times$ objective lens).

Extraction of lysosomal proteins

Lysosome-enrichment protocol was performed according to the manufacturer's instructions (Thermo Fisher Scientific). Briefly, 5×10^7 cells were dissociated using the indicated reagents of the kit and were subjected to ultrasonic treatment. The cell lysate was separated by ultracentrifugation at 145000 $\times$ g and 4°C. After adding Laemmli buffer, samples were analyzed by immunoblotting.

QUANTIFICATION AND STATISTICAL ANALYSIS

All data were analyzed using GraphPad Prism 7.0. All statistical details of the experiments, including exact values of n and what n represents, are mentioned in the figure legends. Statistical significance was determined using the Student's t-test (when comparing two experimental groups), one-way analysis of variance (ANOVA) or two-way ANOVA as appropriate (when comparing more than two experimental groups). p-values < 0.05 were considered statistically significant. Data were expressed as mean $\pm$ standard deviation (SD). ns, not significant; *, p < 0.05; **, p < 0.01; ***, p < 0.001.